Energy
Ville

Network aware market participation for industrial networks

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ATHENS course

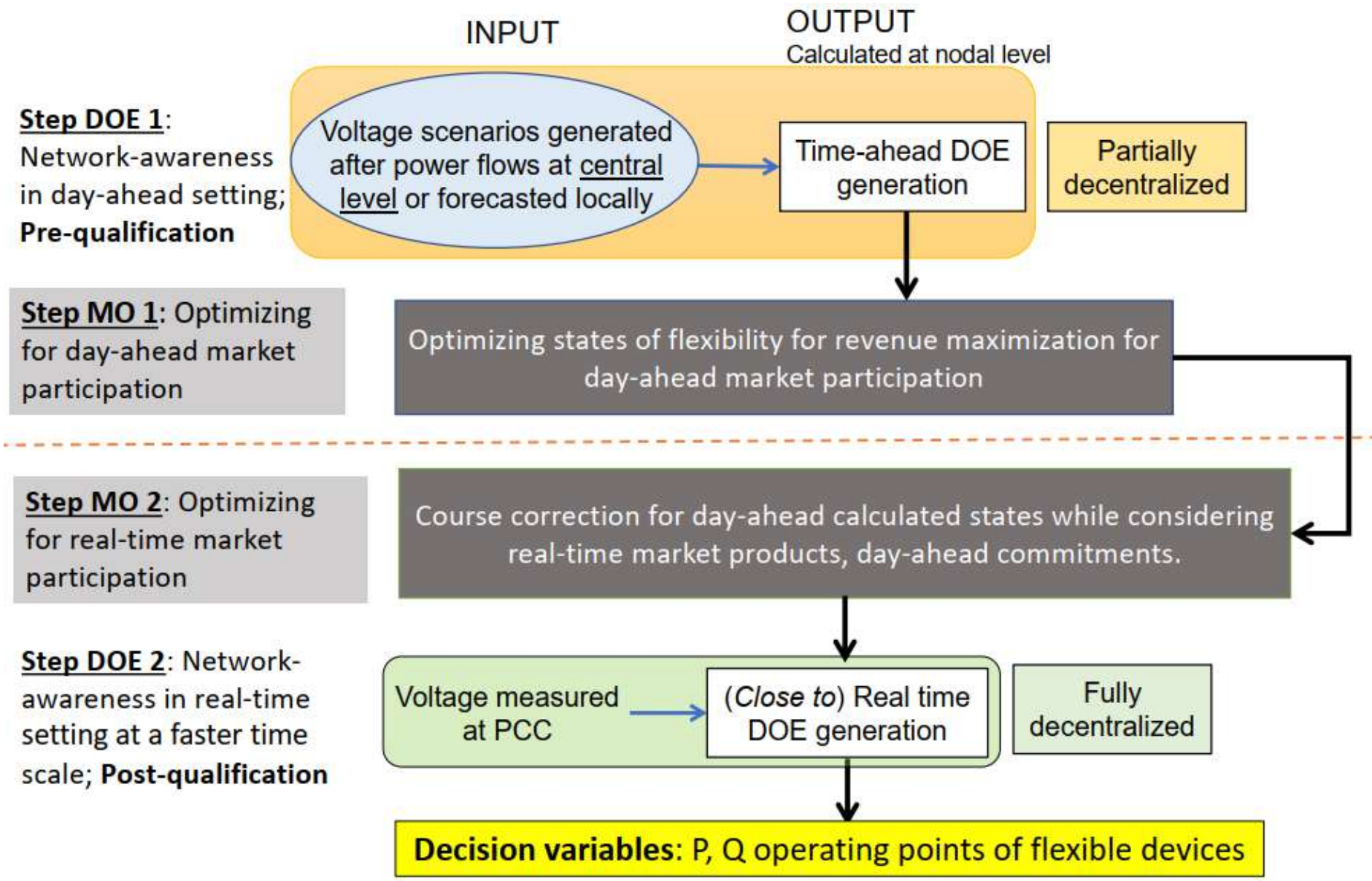
Goal

- Proposing a **two-stage network awareness** framework for utilizing industrial flexibility for energy market participation along with local network awareness
  - First stage: day ahead
  - Second stage: close to real time

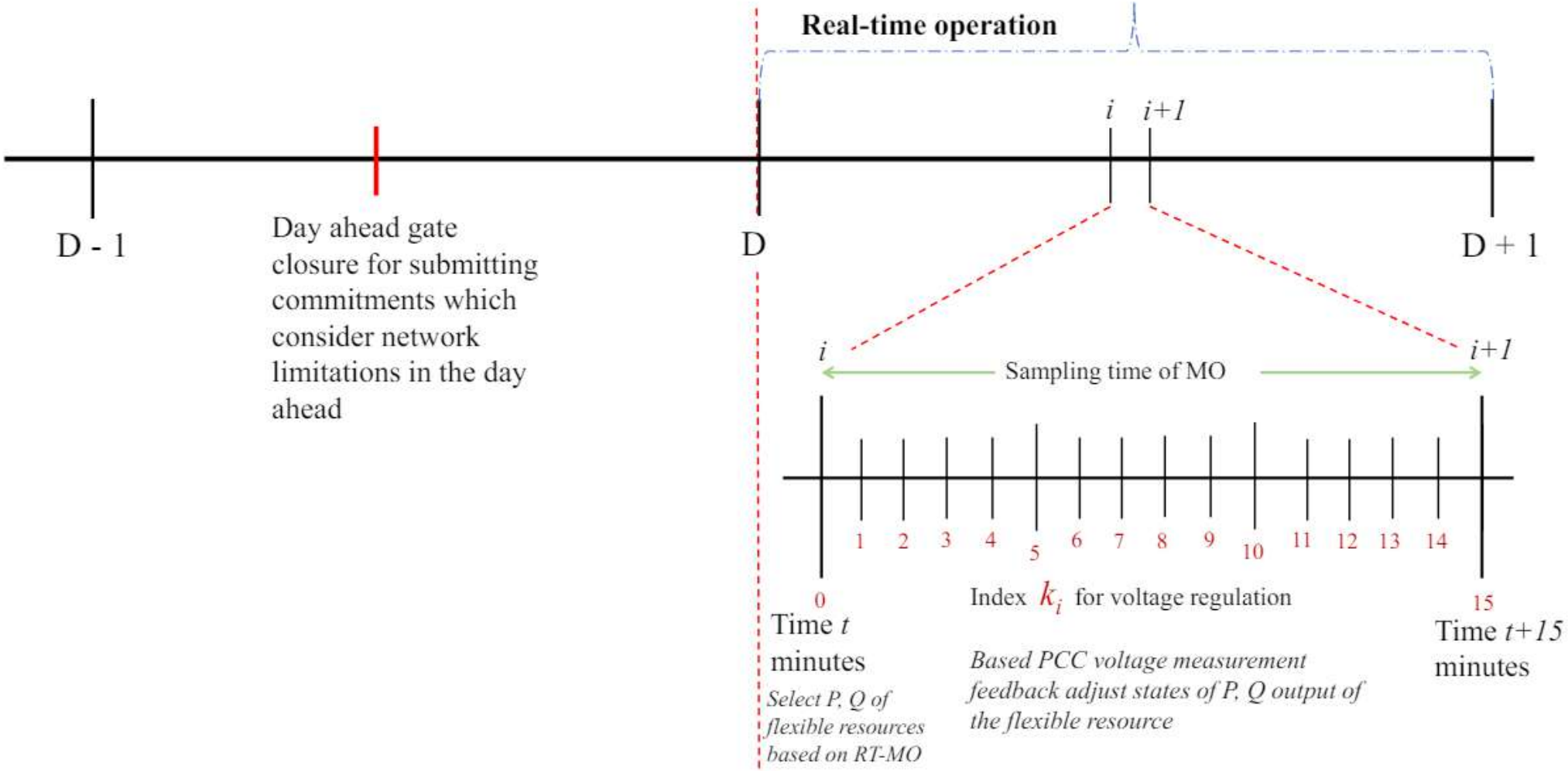
- *Motivation*: Industrial processes are sensitive to fluctuating power quality. Thus, it is expected that network awareness for harnessing industrial flexibility should consider its activation implication on the grid.

Two stage network aware flexibility participation

- Pre-qualifying flexibility states in day-ahead
- Post-qualifying final states of flexibility in close to real time.

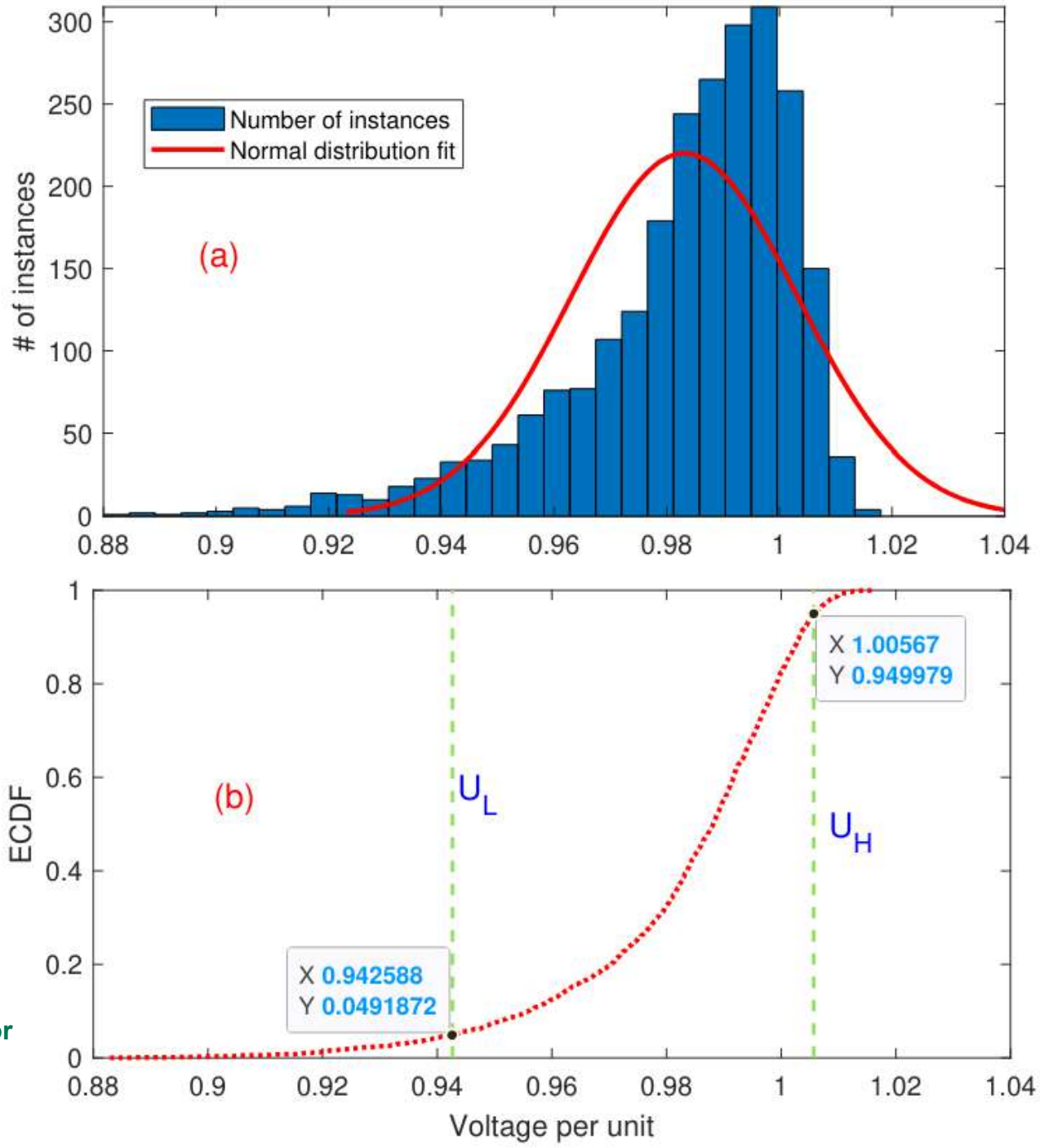


Timeline of operation



Generation of risk averse DOEs

- Risk averse DOEs are calculated based on nodal voltage distribution
- Only the 5% and 95% quartile voltage magnitudes are used for generating the dynamic operating envelopes (DOEs)



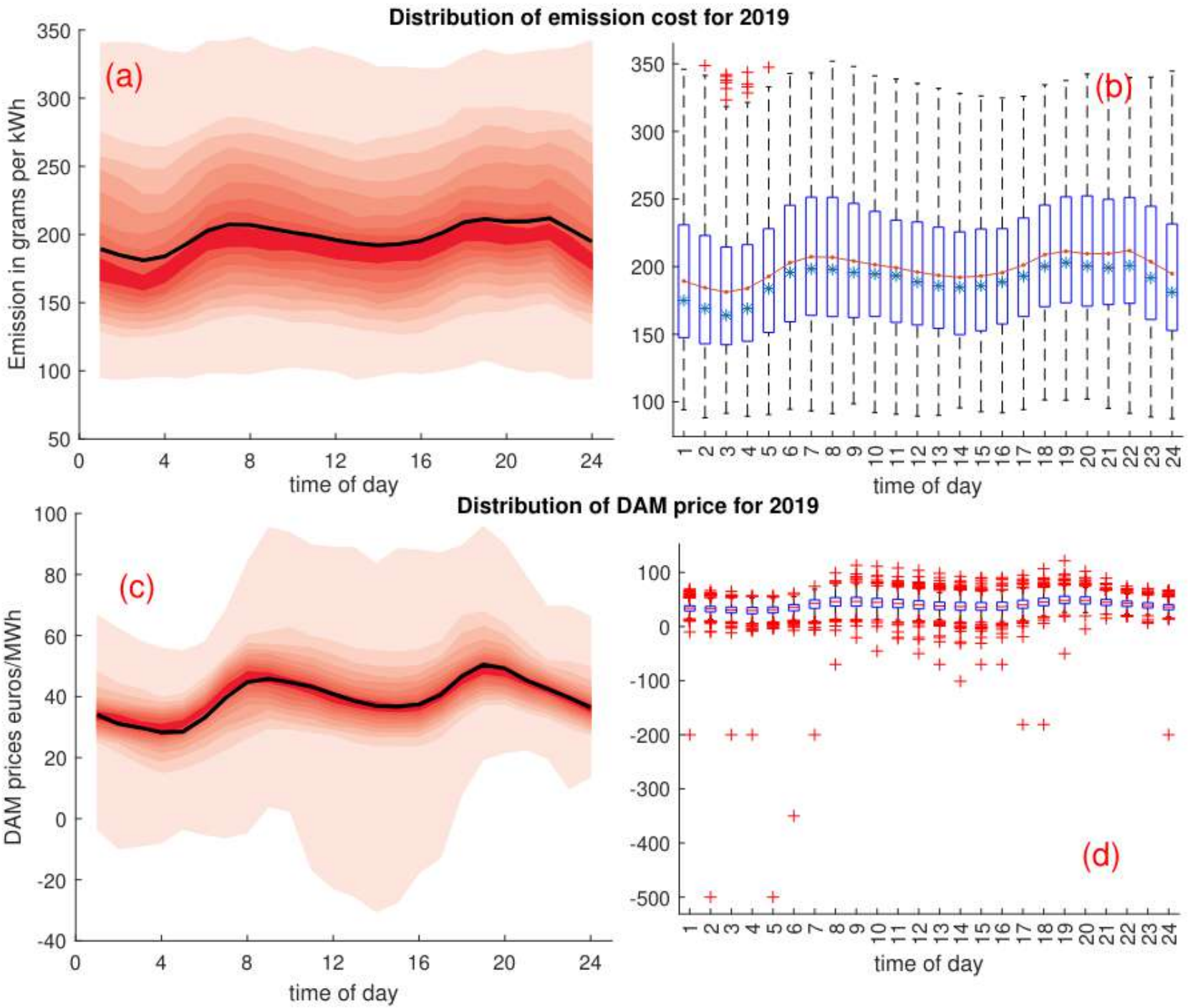
Hashmi, M. U., & Van Hertem, D. (2023). **Robust dynamic operating envelopes for flexibility operation using only local voltage measurement.** *arXiv preprint arXiv:2305.13079*. (at SEST 2023) (Best paper winner)

Market products

- Day ahead market

Objective function:
Sum of the product of day ahead price and flexibility state for the entire day

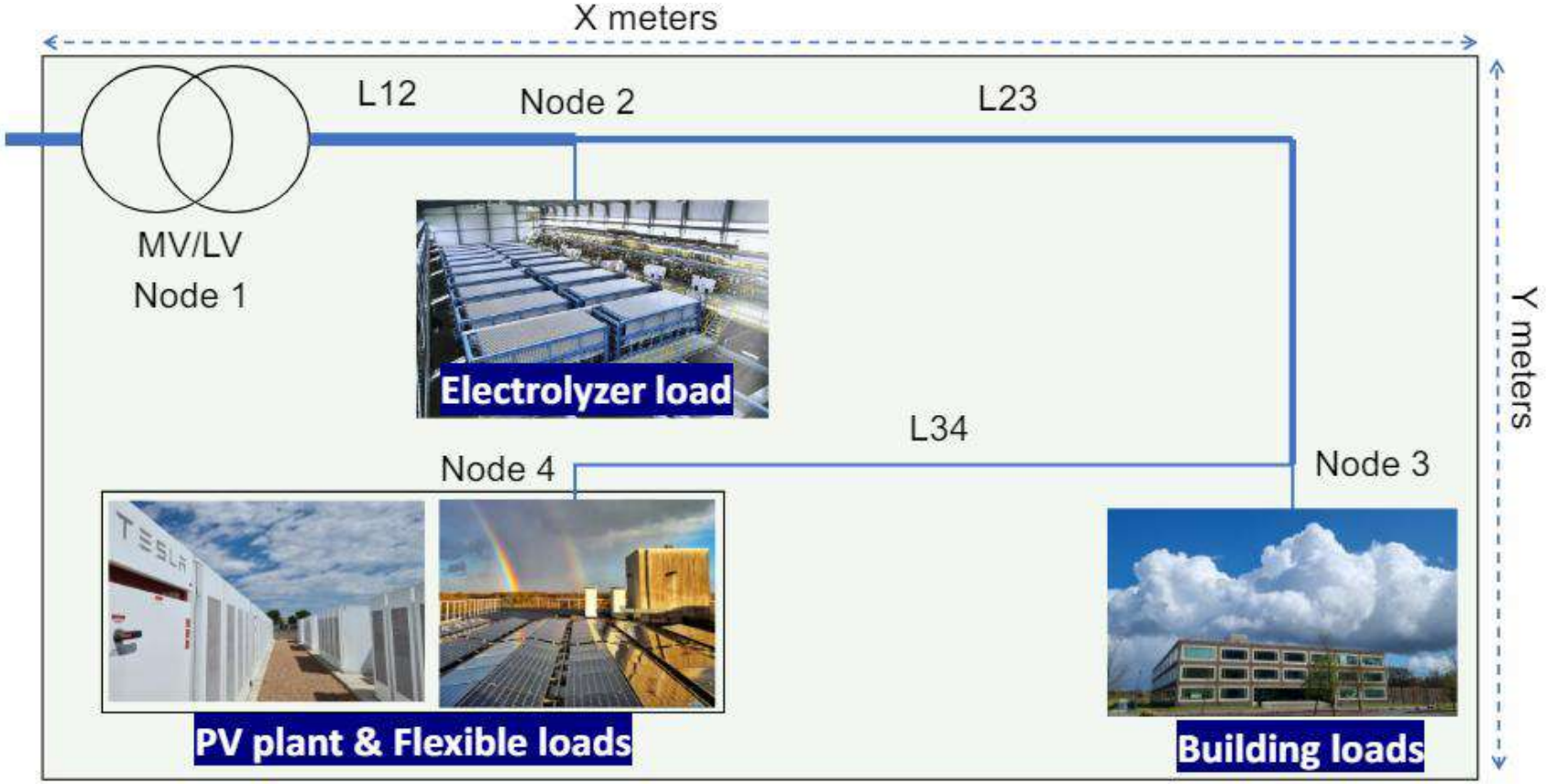
- Emission cost



Stylized industrial network

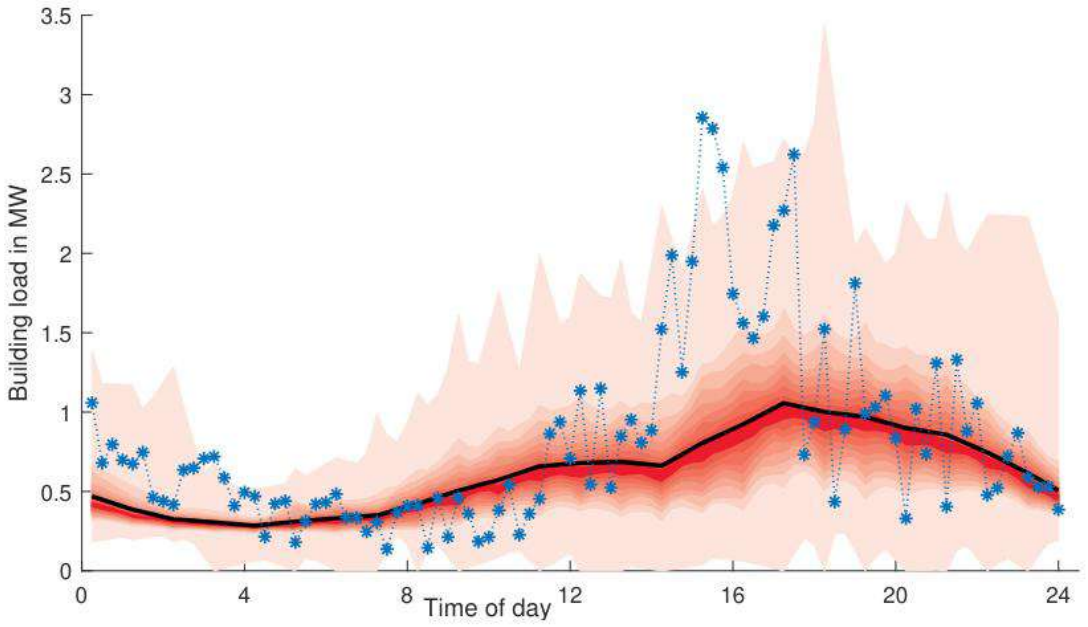
The network consists of (a) industrial process load, (b) building load, (c) PV generation and (d) flexible load.

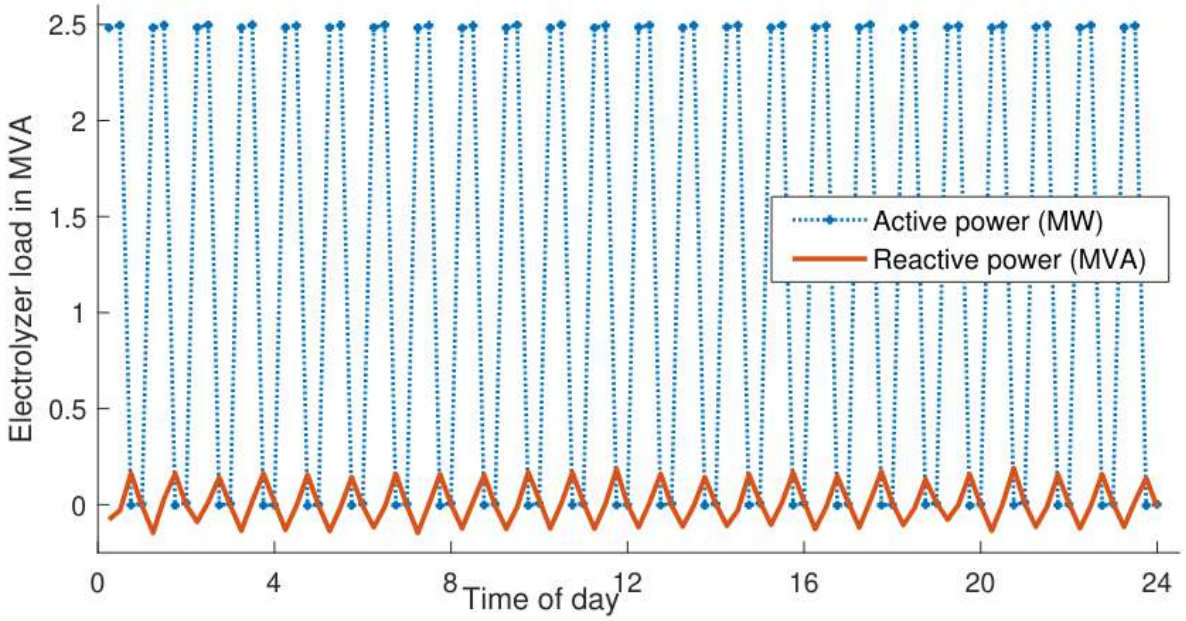
Assumption: the flexible load is modelled as a battery. This may not be true is the flexibility is derived from industrial processes.

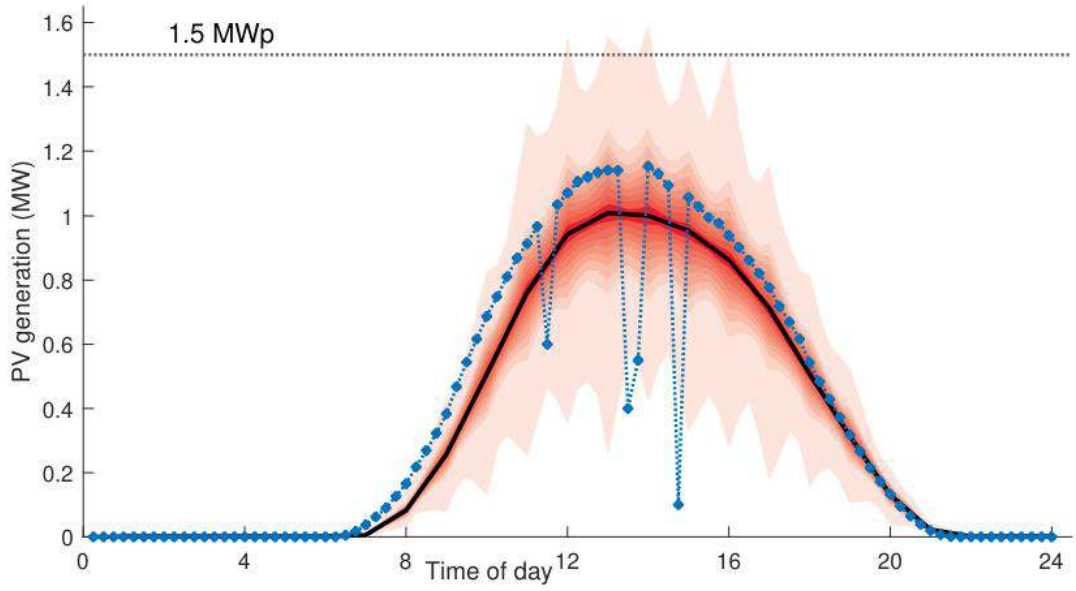


Load & generation profiles

- The electrolyzer load is not considered stochastic
- Building load and PV generation day-ahead distribution and real-time profiles are shown





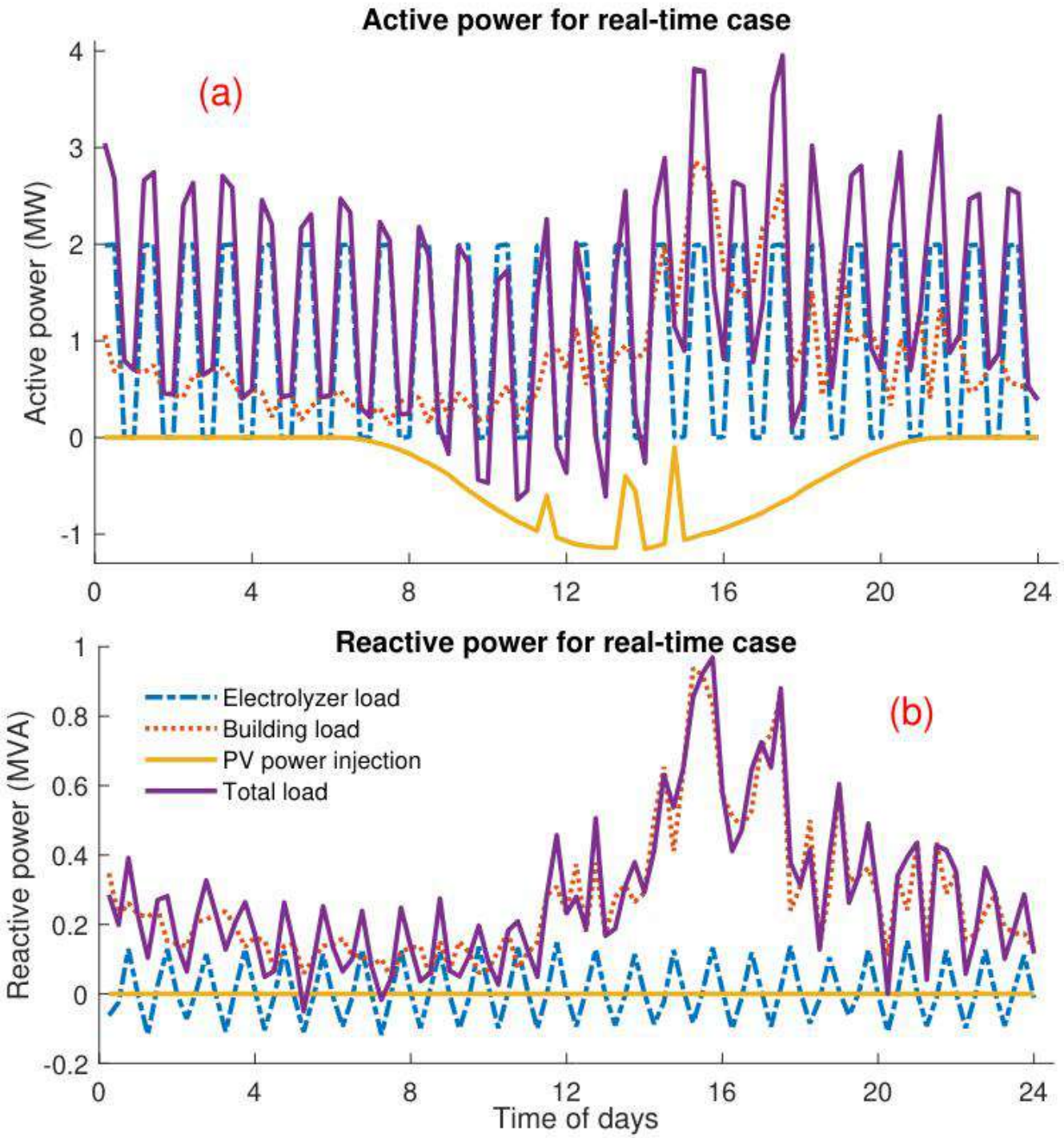


Electrolyzer load profile are provided by Dr Sajid Qazi from Ugent

Net load

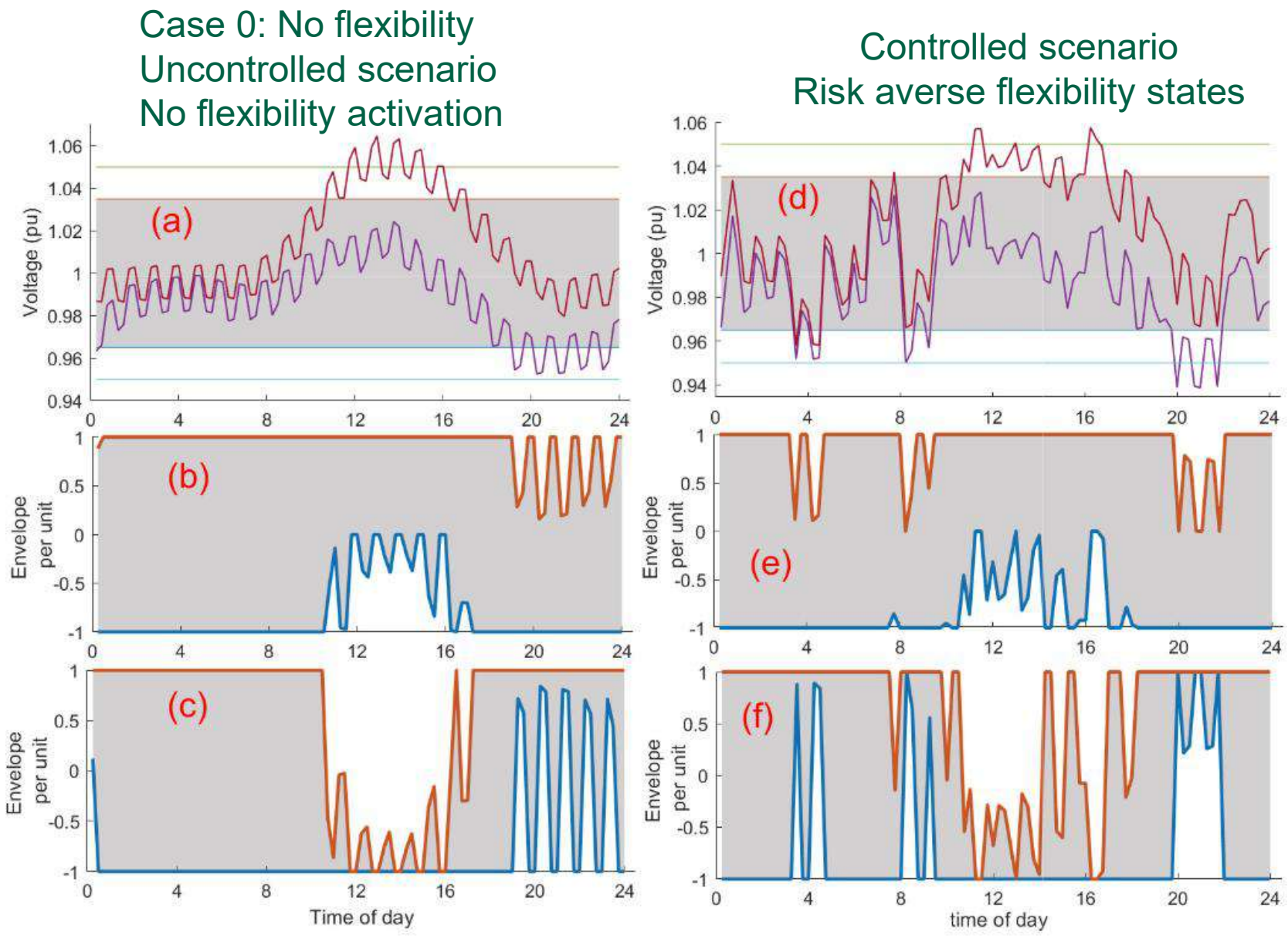
Table 4: Battery parameter

Attribute	Value
$b_{\max}$	1 MWh
$b_{\min}$	0.2 MWh
b_0	0.5 MWh
$\delta_{\max}$	0.5 MW
$\delta_{\min}$	-0.5 MW
η_{ch}	0.95
η_{dis}	0.95
P_B	$\in [-0.5/0.95, .5/0.95]$ MW
Q_B	$\in [-0.5 \times 0.95, .5 \times 0.95]$ MW
$S_{\max}$	0.6 MVA



Day-ahead dynamic operating envelopes

- Activation of flexibility also lead to new network incidents that was not observed



Hashmi, M. U., Deka, D., Bušić, A., & Van Hertem, D. (2022). **Can locational disparity of prosumer energy optimization due to inverter rules be limited?.** *IEEE Transactions on Power Systems*.

KPIs used

$$CVC_{\text{upper}}(U) = \sum_k \mathbf{1}(U > 1 + \Delta_{\text{perm}})(U - 1 - \Delta_{\text{perm}}),$$

$$CVC_{\text{lower}}(U) = \sum_k \mathbf{1}(U < 1 - \Delta_{\text{perm}})(1 - \Delta_{\text{perm}} - U).$$

- KPI 1: Market participation revenue
- KPI 2: Battery cycles of operation
- KPI 3: Cumulative voltage correction
- KPI 4: Voltage correction index
- KPI 5: emission cost
- KPI 6: power factor metrics
- KPI 7: Load parameters (peaks, dips, variance)

$$VCI_1(U) = \sum_k \mathbf{1}(U > U_{\max}),$$

$$VCI_2(U) = \sum_k \mathbf{1}(U > 1 + \Delta_{\text{perm}}),$$

$$VCI_3(U) = \sum_k \mathbf{1}(U < 1 - \Delta_{\text{perm}}),$$

$$VCI_4(U) = \sum_k \mathbf{1}(U < U_{\min}).$$

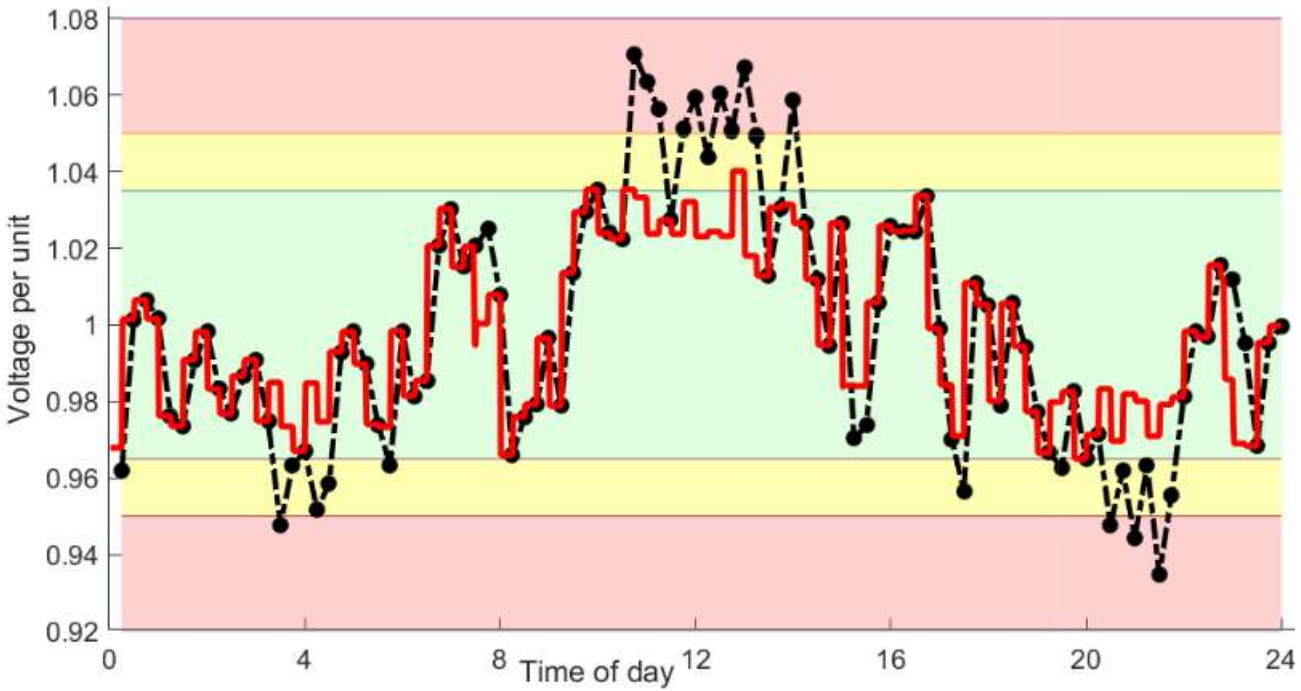
Case evaluation

- **Case 0**: No flexibility considered
- **Case 1**: Industrial flexibility energy market participation with no network consideration,
- **Case 2**: Industrial flexibility energy market participation with DA network consideration,
- **Case 3**: Industrial flexibility energy market participation with RT network consideration, and
- **Case 4**: Industrial flexibility energy market participation with DA and RT network consideration

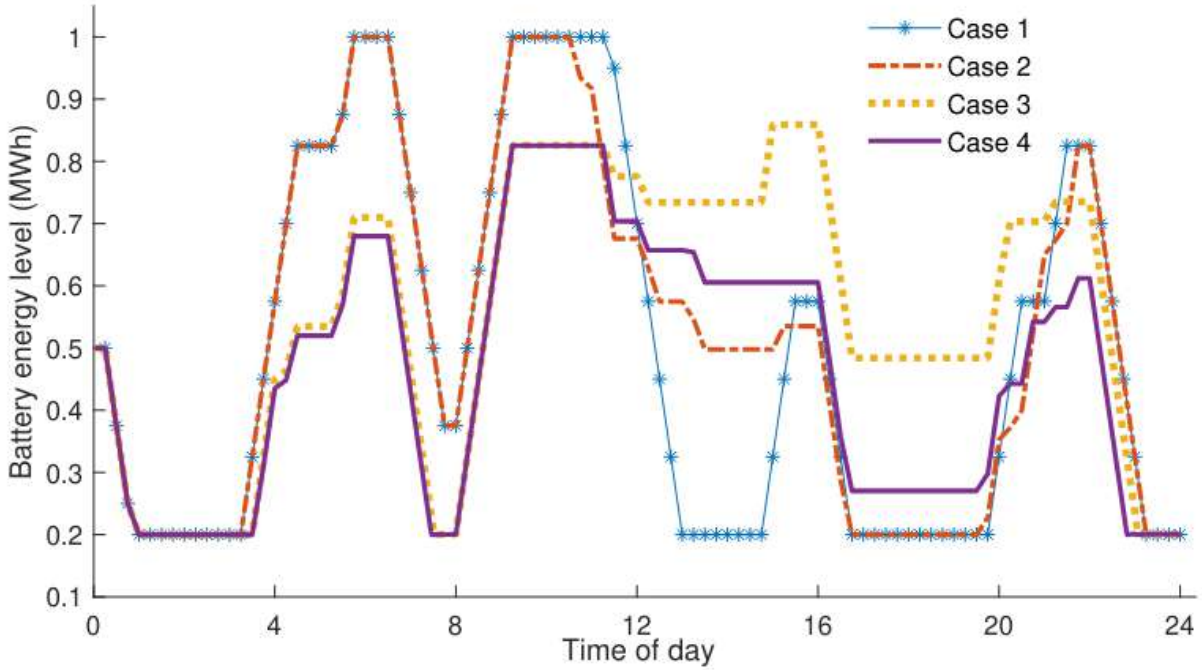
KPI	Description	Case 0	Case 1	Case 2	Case 3	Case 4
KPI1 Profit	cost reduction	0	57.18	48.21	32.39	34.93
KPI2 cycles operation	100% DoD	0	2.44	2.0931	1.523	1.456
KPI3 CVC	CVC_{upper}	0.0828	0.2073	0.2473	0.221	0.088
	CVC_{lower}	0.1309	0.3427	0.2695	1.276	1.18
KPI4 VCI Node4	$\sum VCI_1$	0.0167	0.0833	0.15	0	0
	$\sum VCI_2$	0.167	0.1833	0.2	1.75	0.75
	$\sum VCI_3$	0.033	0.1667	0.2333	0.25	0
	$\sum VCI_4$	0	0.0833	0.0667	0	0
KPI4 VCI	$\sum VCI_1$	0.25	1.25	2.25	0	0
	$\sum VCI_2$	2.5	2.75	3	1.75	0.75
	$\sum VCI_3$	2.5	5	6.25	2.25	1.5
	$\sum VCI_4$	1	2.5	2.25	0.75	0.75
KPI5 emission cost	Emission cost	8.76E+03	8.73E+03	8.72E+03	8.67E+03	8.67E+03
	Emission saving	0	36.27	44.92	92.82	95.54
KPI6 PF metrics	Mean |PF|	0.91	0.9199	0.889	0.887	0.871
	$\sum 1(\lvert PF \rvert < 0.8)$	0.1146	0.125	0.198	0.167	0.218
KPI7: Load parameters	P peak	3.954	4.340	3.970	3.954	3.954
	P min	-0.643	-1.087	-0.891	-0.643	-0.643
	P var	1.359	1.592	1.408	1.344	1.320
	Q peak	0.969	0.969	1.016	0.873	1.106
	Q min	-0.050	-0.050	-0.469	-0.208	-0.263
	Q var	0.046	0.046	0.112	0.0523	0.091
	S peak	4.050	4.423	4.050	3.964	3.964
	S min	0.180	0.274	0.095	0.274	0.274
	S var	1.064	1.147	1.039	0.967	0.898

Real time network awareness

Voltage correction compared to Case 1

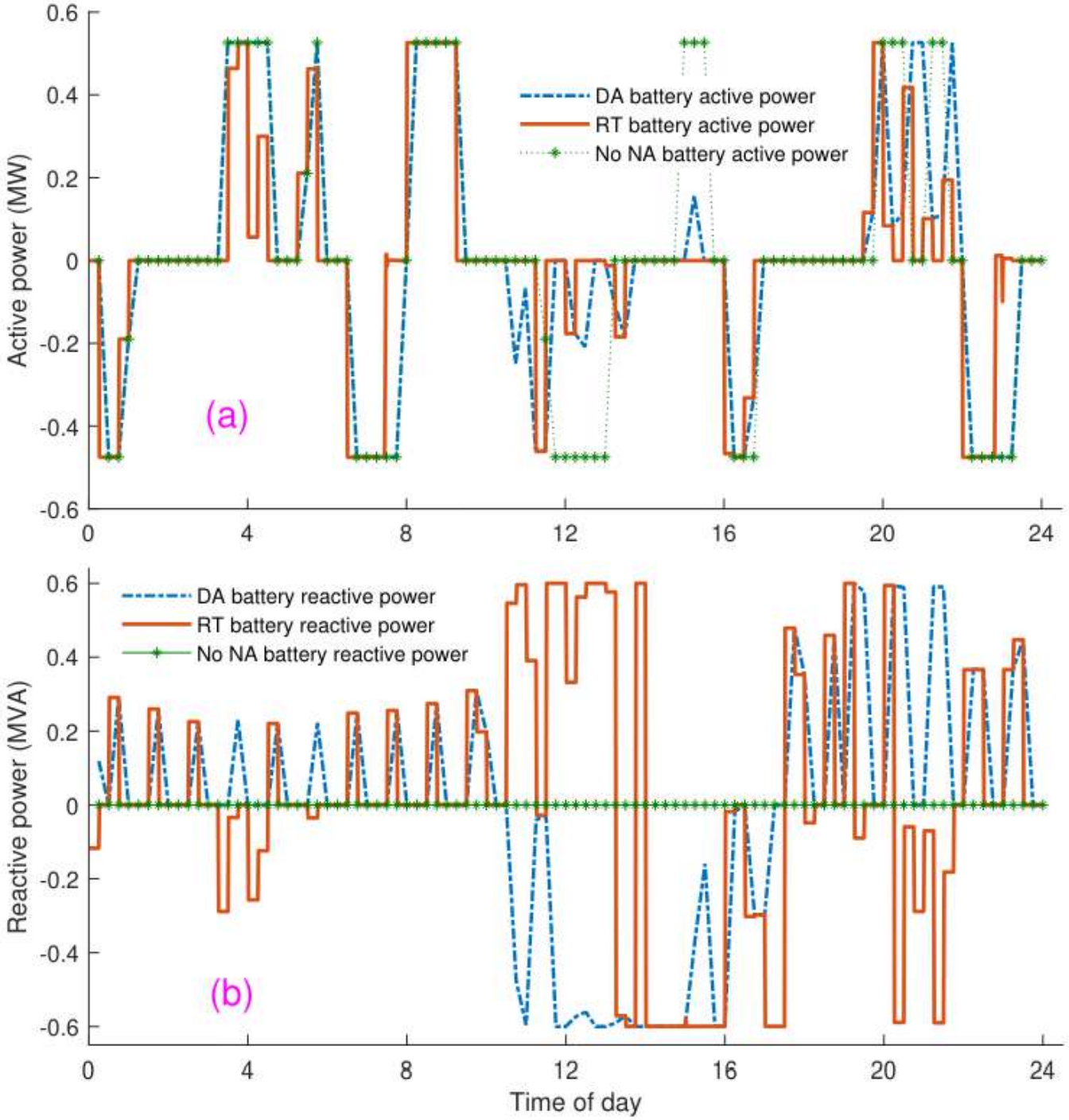


Evolution of battery capacity for different cases



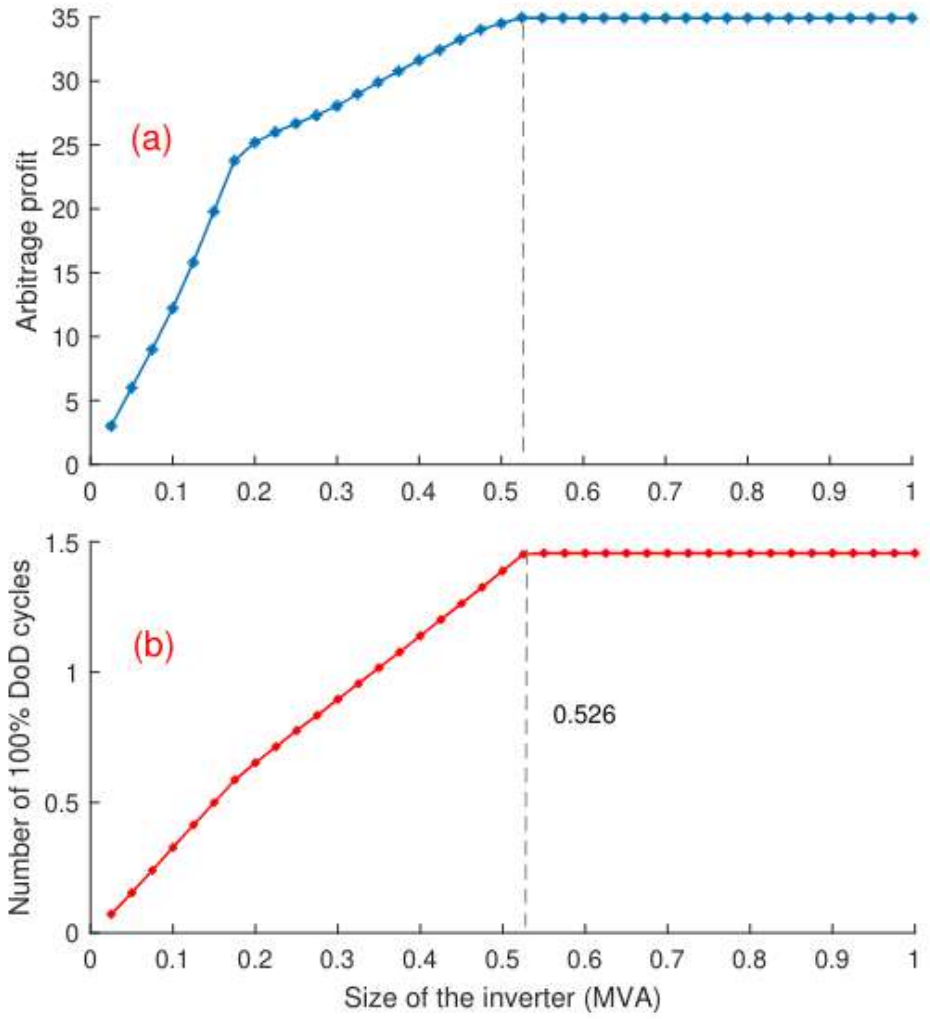
Importance of Q

- Reactive power control is crucial to ensure that active power deviation due to network awareness is minimized

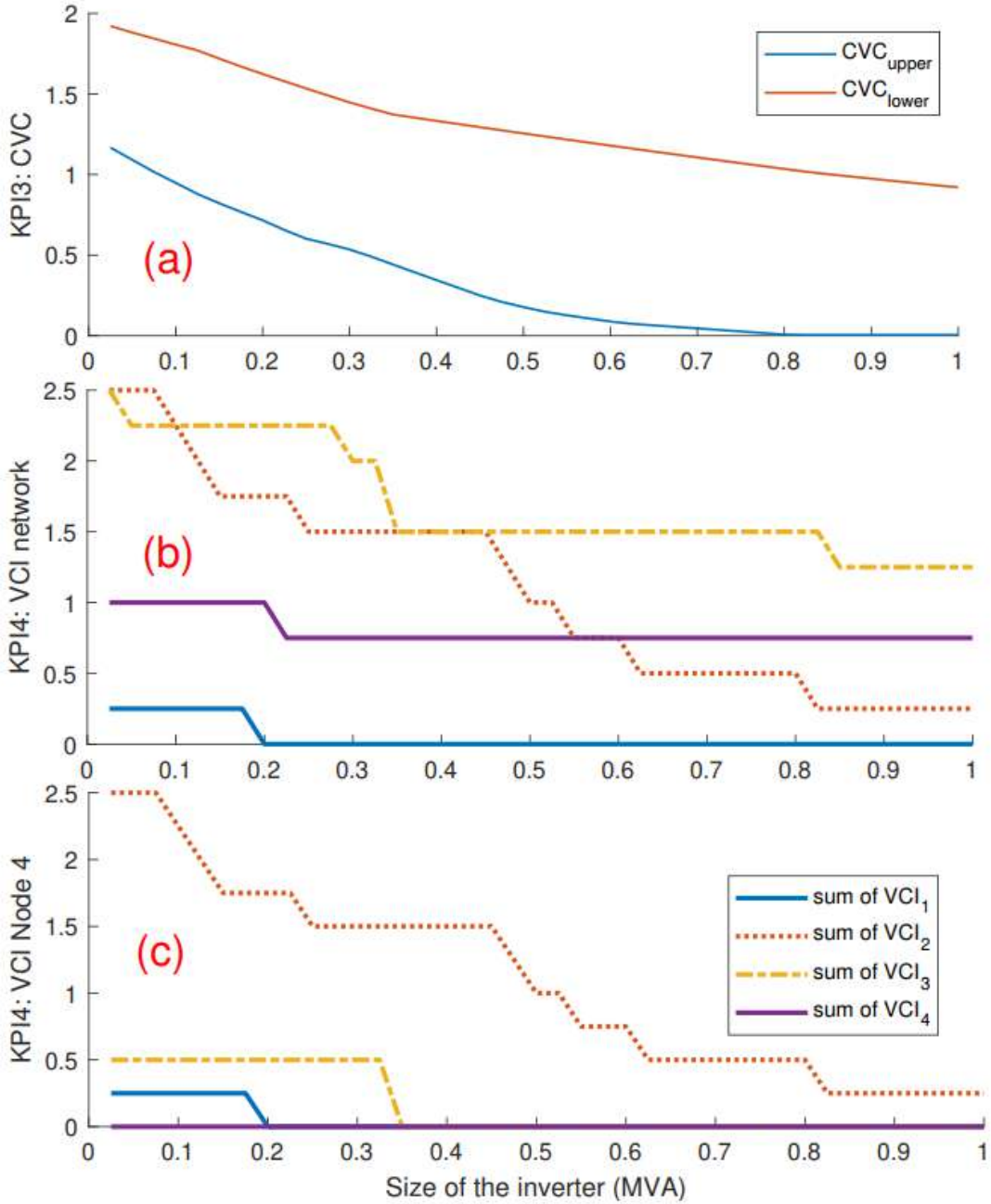


Sensitivity of inverter sizing

Implication on active power gains



Implication on Voltage correction



Key Conclusions

- We presented **two-stage network awareness framework** for utilizing flexible resources ensuring activation of such flexibility not only avoids creation of any new network issues but also actively solves pre-existing network issues.
- Network awareness lead to up to 39% reduction in the market participation revenue but voltage correction index and cumulative voltage correction improved substantially
- Also performed sensitivity analysis for inverter sizing for showing the implication of inverter size on active power profit and voltage correction.

Thank you for your attention!

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